

Stable but Unsafe

Gauge-compatible nonmodal amplification, controller sweeps, cycle pressure, and stochastic barriers in an IEEE 39-bus reduced surrogate

Status: mechanism-level numerical campaign, 23 August 2026. This report is deliberately not presented as full ANDES/EMT validation.

1. What was actually simulated

I rebuilt an independent AC-consistent reduced IEEE 39-bus study from the MATPOWER case39 static network. AC power flow is re-solved for load and topology changes, the synchronizing Jacobian is Kron-reduced to generator ports, and SG/GFL/GFM reduced dynamics are attached. The GFL block follows the PI-PLL structure used in the scalarizability manuscript. A controller-network coupling scale γ multiplies the reduced active-power feedback channel; γ is a mechanism/stress parameter, not a manufacturer PLL setting. Because the companion reproducibility source code was not available in the uploaded files, the results below are an independent falsification-oriented campaign rather than a numerical reproduction of earlier manuscript values.

The physical transient metric initializes only network angle/frequency coordinates and measures only network electromechanical energy $E = \theta^T K \theta / (2 \omega_b) + \omega^T M \omega / 2$. Controller states start at equilibrium and are excluded from the energy output. This avoids assigning an arbitrary Euclidean energy to PLL integrator states. The maximum gain is $G_{\text{phys}} = \sup_t \sigma_{\max}^2(C_E \exp(A t) B_E)$.

2. Main result: a modal-safe system can still be strongly amplifying

Case	Kp	gamma	critical pole	zeta_min	G_phys	max t_peak	F_0	F_inf	effective rank
Low amplification	46	0.4	-0.7075 + j12.1133	5.106%	2.41	22.4 ms	-0.051	0.879	15.47
High amplification	60	2.0	-0.6964 + j12.1245	5.117%	10.94	13.8 ms	-0.057	2.393	5.85

The two highlighted cases have almost the same dominant small-signal information: the spectral abscissae differ by only 1.57%, the critical frequencies by 0.09%, and $\zeta_{\min}$ by 0.22%. Yet the worst physical transient energy gain differs by 4.54x. This is the cleanest mechanism-level evidence obtained in this campaign that dominant poles alone do not order pre-asymptotic physical amplification.

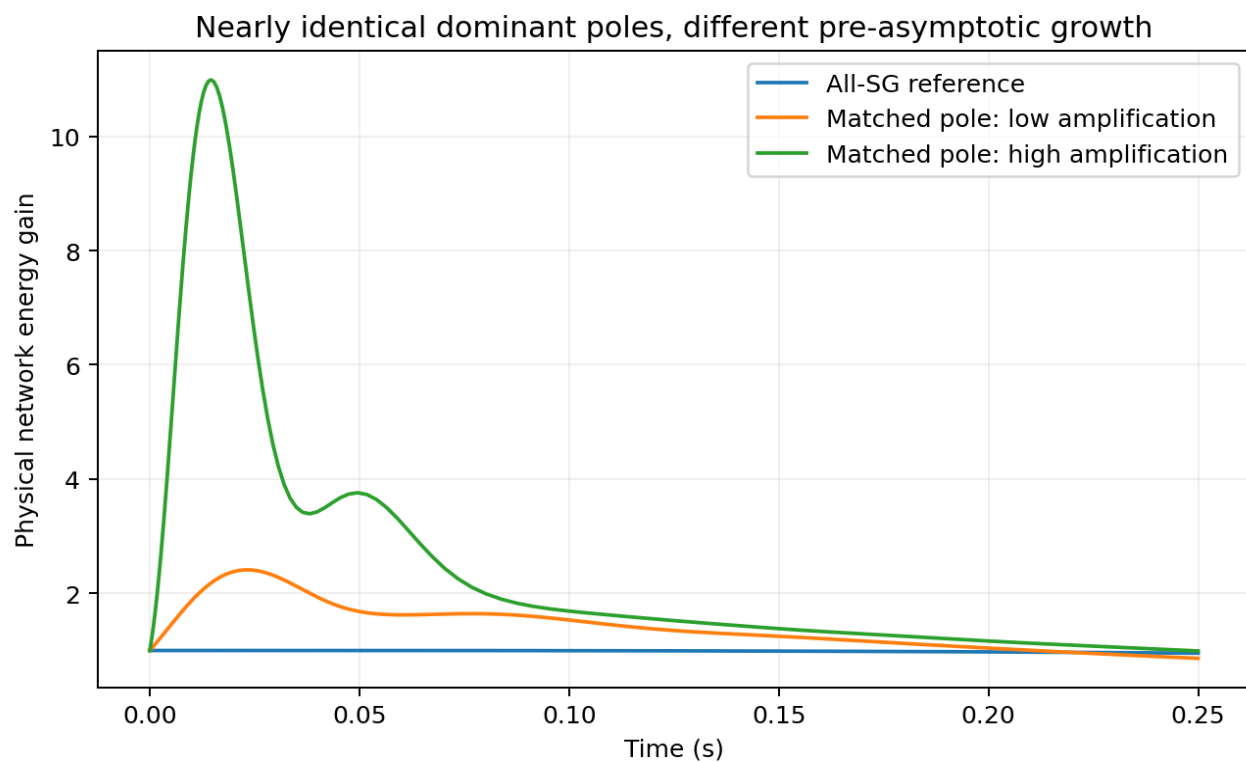


Fig. 1. Transient physical-energy gain for an all-SG reference and two cases with nearly matched dominant poles.

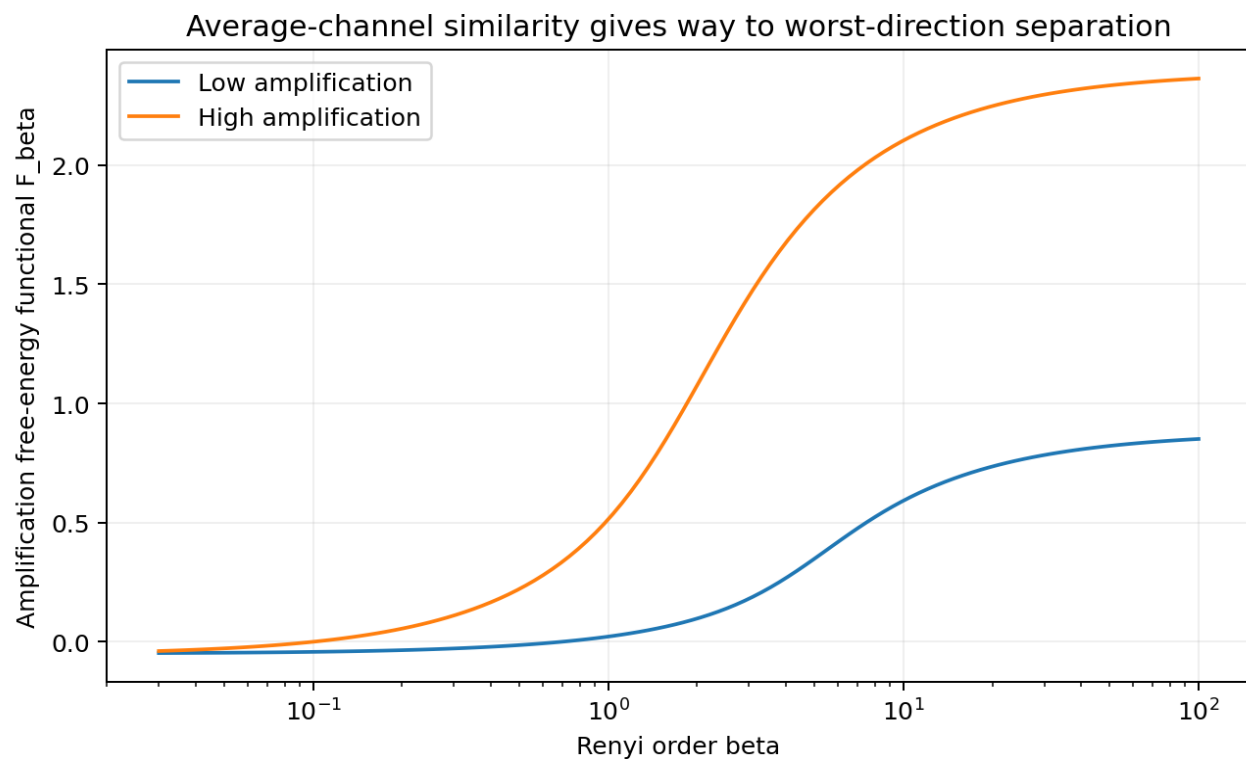


Fig. 2. Renyi-order amplification functional. The $\beta \rightarrow 0$ geometric-average behavior is similar, while large β exposes concentration into a few dangerous singular directions.

3. Dense controller sweeps

The Kp-Ki campaign contains 775 points. All are asymptotically stable in this selected mechanism range. 614 points satisfy $\text{zeta_min} \geq 5\%$; every one of those modal-safe points has $G_{\text{phys}} > 5$, with G_{phys} ranging from 5.73 to 10.44. The broader gamma-Kp campaign contains 841 points, of which 736 are modal-safe. Within that modal-safe region G_{phys} ranges from 1.49 to 32.20.

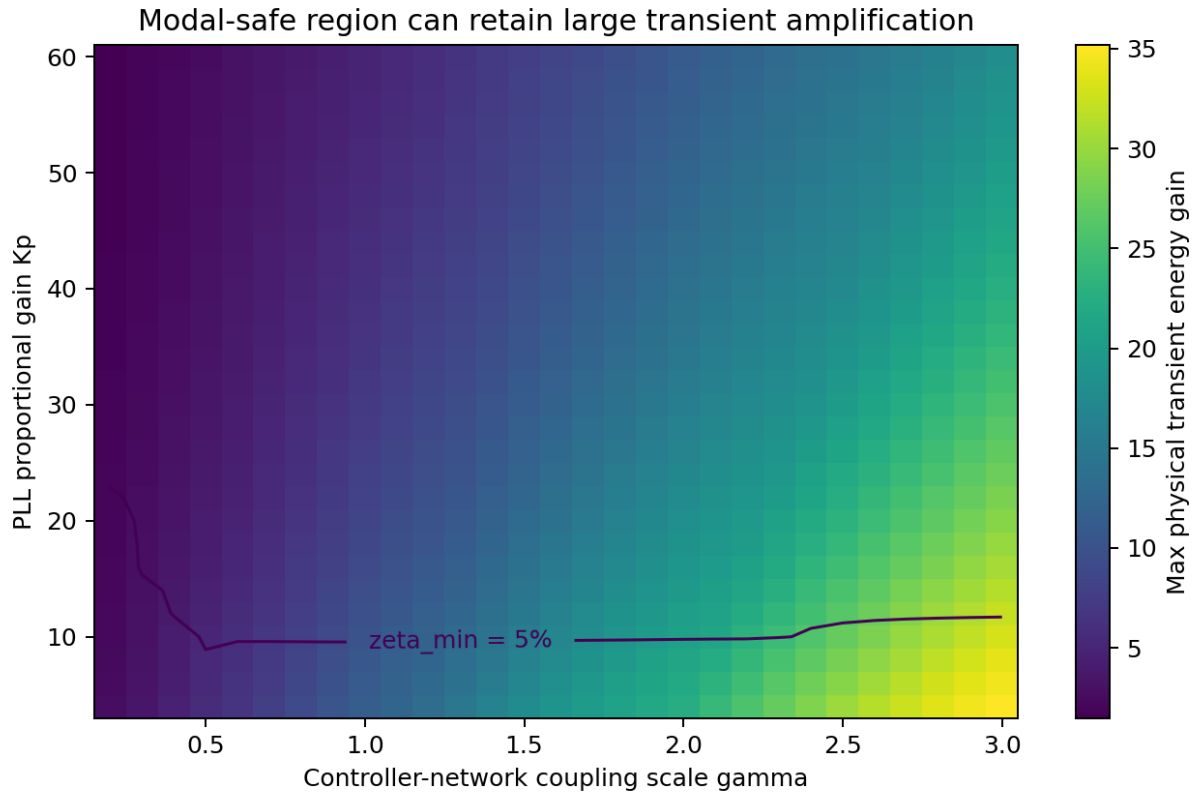


Fig. 3. Controller gain and controller-network coupling can change transient amplification by more than an order of magnitude while the modal damping screen remains above 5%.

The mechanism is technology- and coupling-dependent, not a claim that every well-damped GFL system is hidden-unsafe. The useful result is existence plus a reproducible parameter region: a modal damping target is not sufficient to rank transient amplification once controller-network feedback is non-normal.

4. Penetration and technology negative control

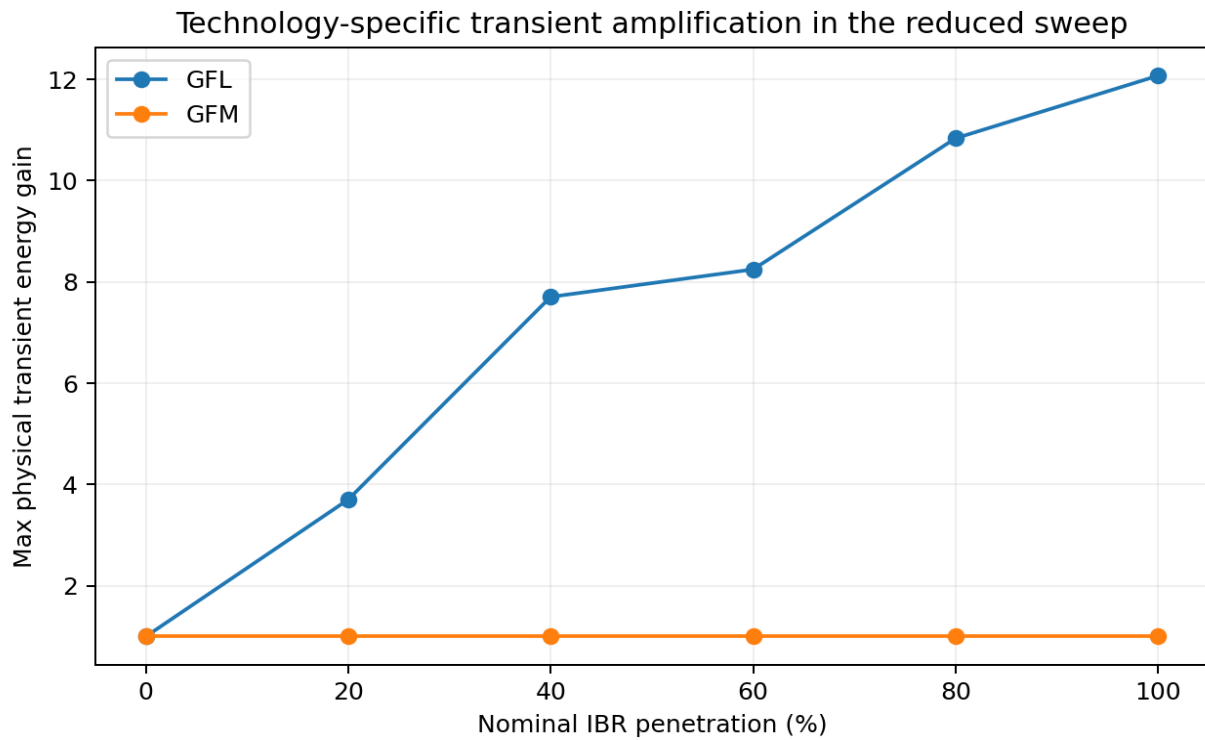


Fig. 4. Under the tested surrogates, the GFL family develops strong physical transient gain with penetration, while the passive lagged-droop GFM helper remains contractive in the chosen physical energy metric.

Technology	Penetration	zeta_min	G_phys
GFL	0%	5.11%	1.00
GFL	20%	5.12%	3.70
GFL	40%	5.13%	7.71
GFL	60%	5.13%	8.25
GFL	80%	11.11%	10.84
GFL	100%	11.31%	12.07
GFM	0%	5.11%	1.00
GFM	20%	3.86%	1.00
GFM	40%	3.38%	1.00
GFM	60%	3.38%	1.00
GFM	80%	3.23%	1.00
GFM	100%	3.19%	1.00

The exact GFM value $G_{\text{phys}} = 1$ in this campaign is a property of the deliberately passive first-order droop helper used here; it is a negative control, not a universal statement about grid-forming converters.

5. Topology and operating point

A standardized new line 25-38, using the existing 25-26 branch as the impedance template, increases standard Fiedler connectivity by 0.345% and GFV by 0.230% at $K_p=25$. In this independent surrogate it also reduces G_{phys} by 8.94%. Thus this campaign does not produce a topology-induced nonmodal Braess reversal; instead it gives a useful negative result: reinforcement helps, but the residual transient gain remains large.

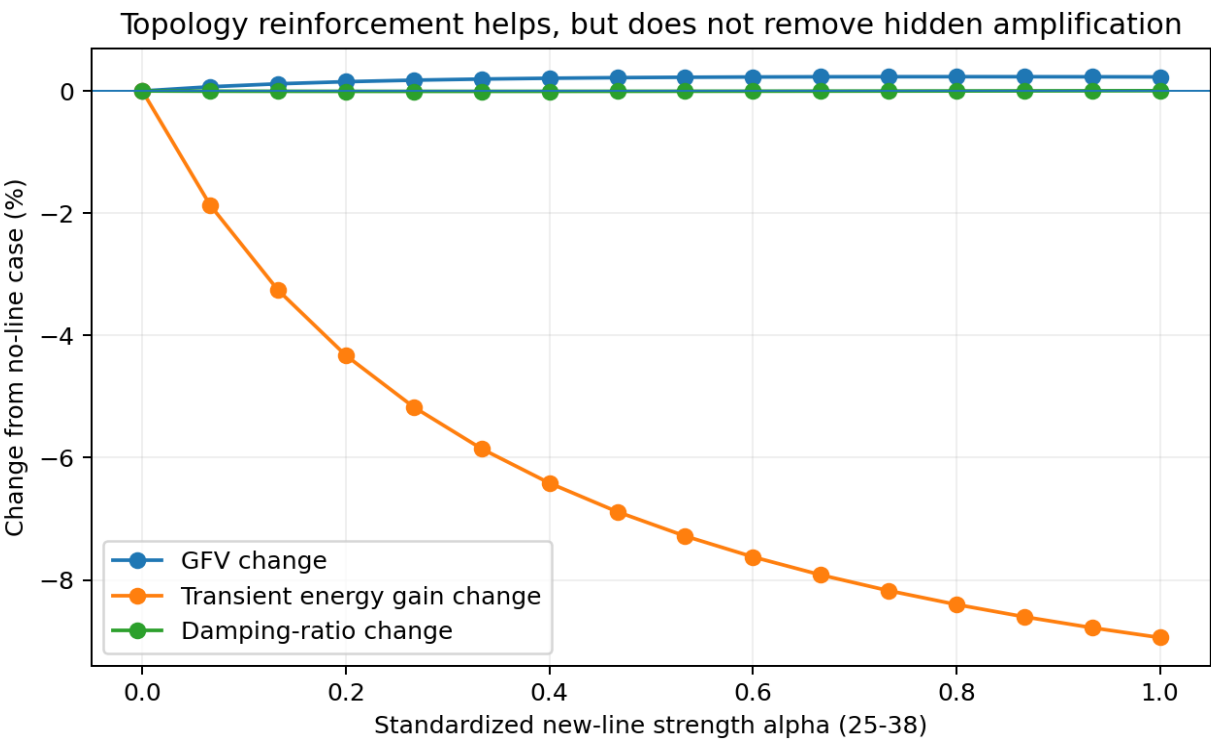


Fig. 5. Standardized line-strength sweep. Connectivity rises and transient gain falls, but the topology action does not eliminate the controller-created amplification.

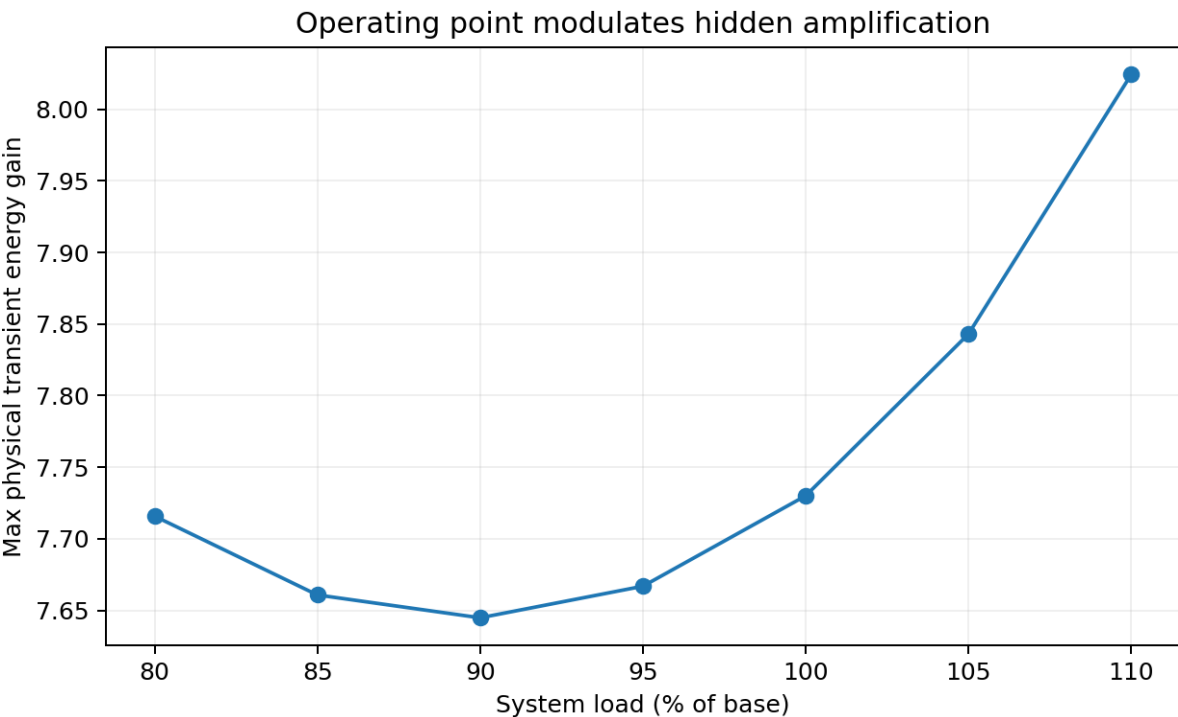


Fig. 6. Load changes from 80% to 110% modulate the transient gain but are a secondary effect relative to controller coupling in this model.

6. Cycle pressure and the limits of the isolated-mode scaling law

I formed an intervention-cycle matrix Q from unit local damping actions at the binding frequency. This is an action-space diagnostic derived from the physical frequency-response matrix; it is not identical to the Q used in the earlier rank-one companion scaling experiment. Co-retiring inertia and damping increases the binding-frequency cycle radius, while preserving absolute damping largely suppresses that growth. However, tracking one eigenmode continuously gives a co-retirement log-log slope -1.161 , not $-1/2$, and a fixed- D slope -0.011 . Therefore the earlier isolated-mode $\alpha^{(-1/2)}$ Q -norm law does not quantitatively carry over to this multimode controller-aware pressure object.

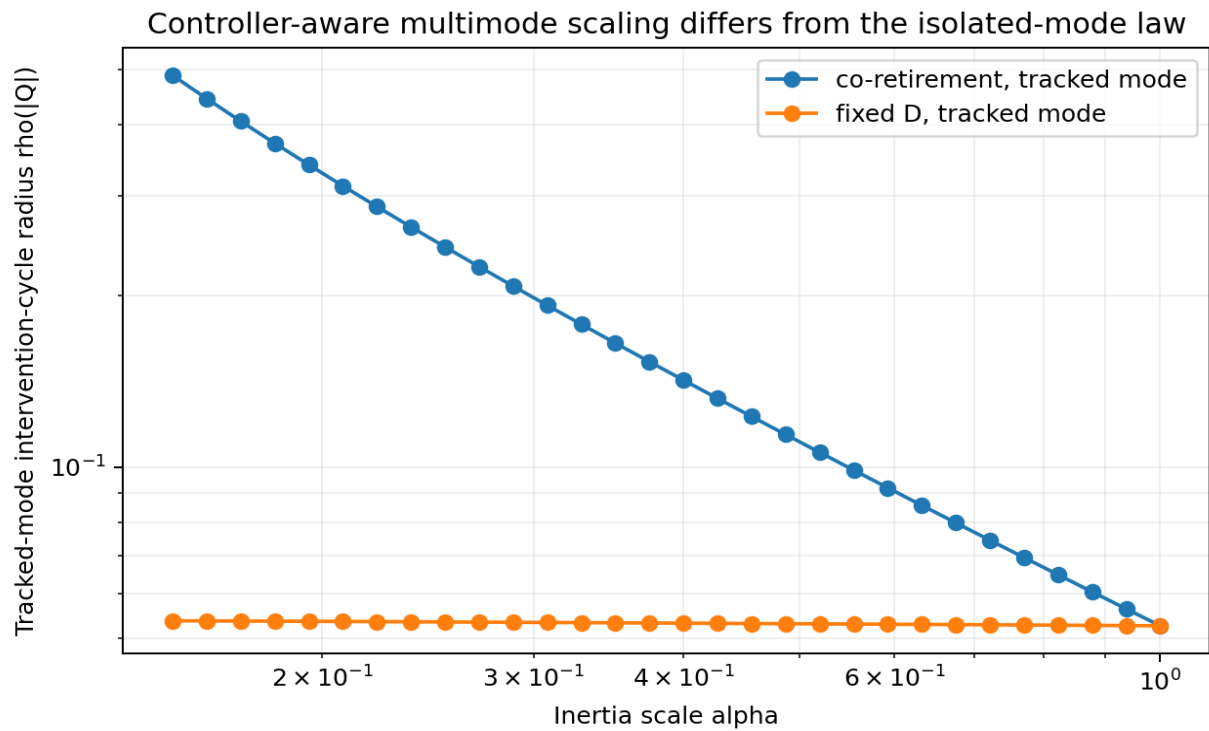


Fig. 7. Tracked-mode interaction-cycle radius. The qualitative co-retirement amplification survives, but the exponent is model/object dependent.

7. A localized-to-collective interaction switch

A dense co-retirement scan around $\alpha = 0.273$ reveals a binding-mode switch. Between $\alpha=0.272$ and 0.273 the spectral abscissa, $\zeta_{\min}$, and G_{phys} are smooth, but the binding frequency jumps from 23.27 to 4.66 rad/s. At the same point the effective number of pair-cycle contributions jumps from 2.34 to 29.54 and the cycle entropy from 0.94 to 2.03. This should not be called a thermodynamic phase transition; it is a mode-family switch accompanied by a sharp change in the interaction topology seen by the cycle diagnostic.

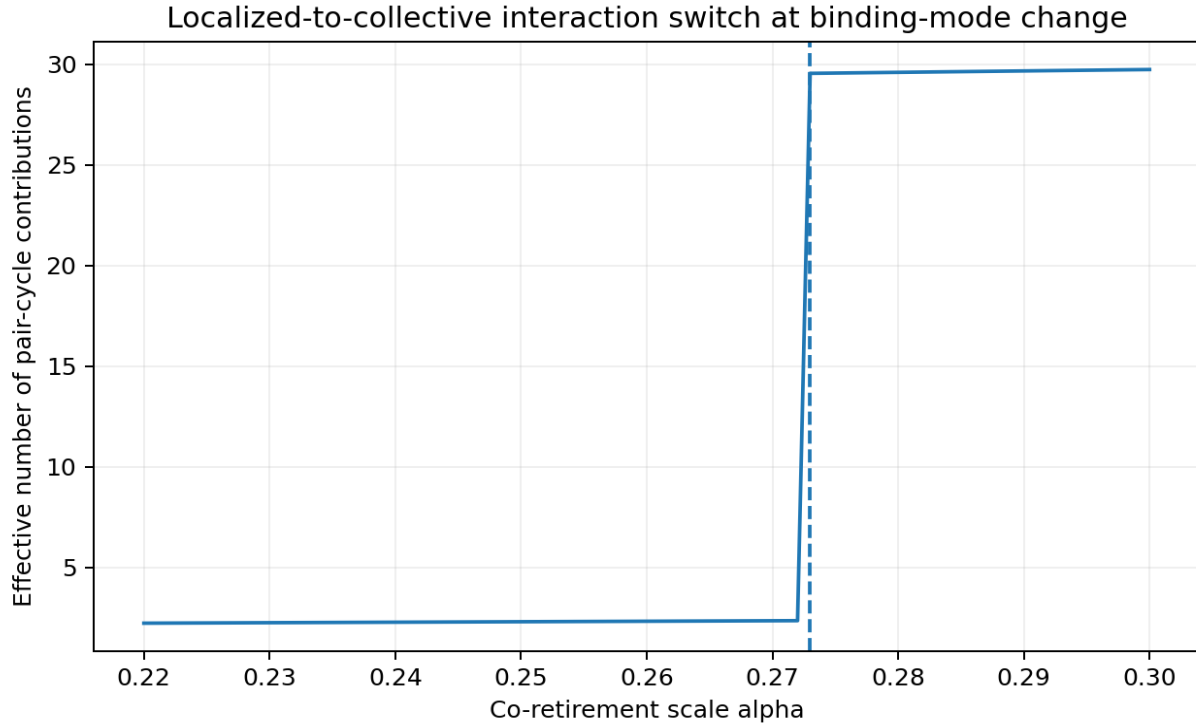


Fig. 8. The interaction structure changes from a few dominant pair cycles to a broad collective feedback set.

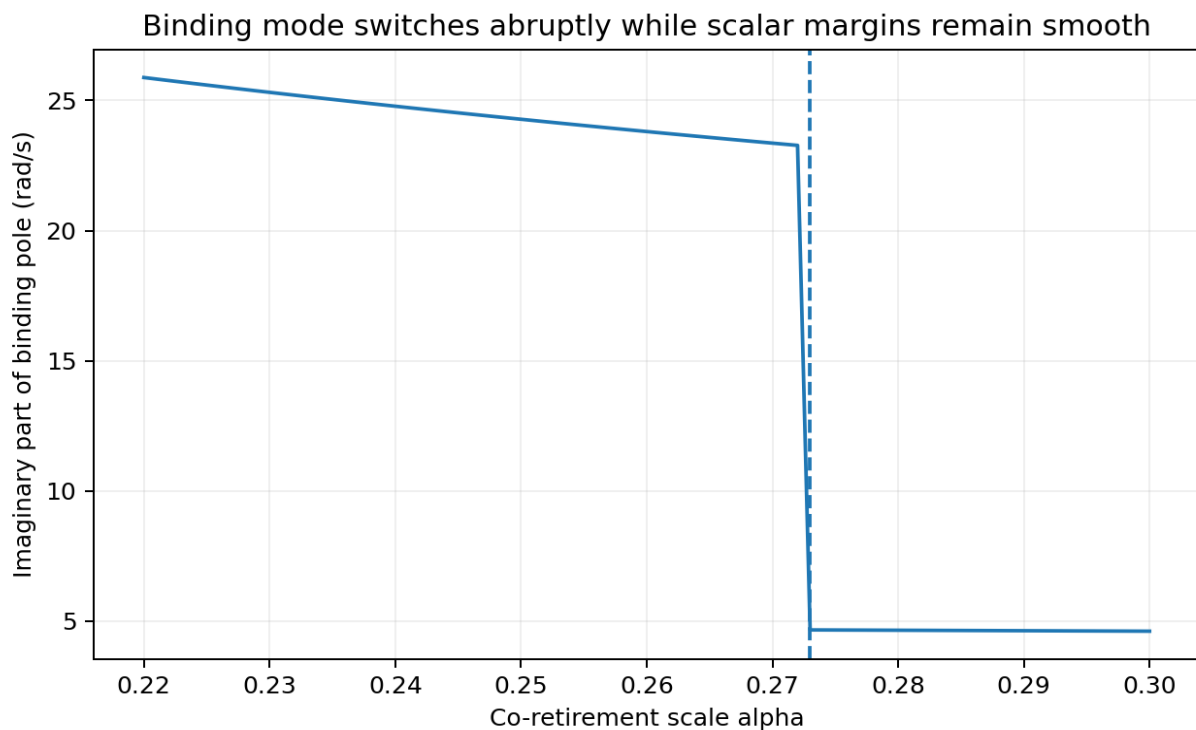


Fig. 9. The switch is tied to a change in the binding mode family.

8. Virtual inertia/damping sweep for the GFM helper

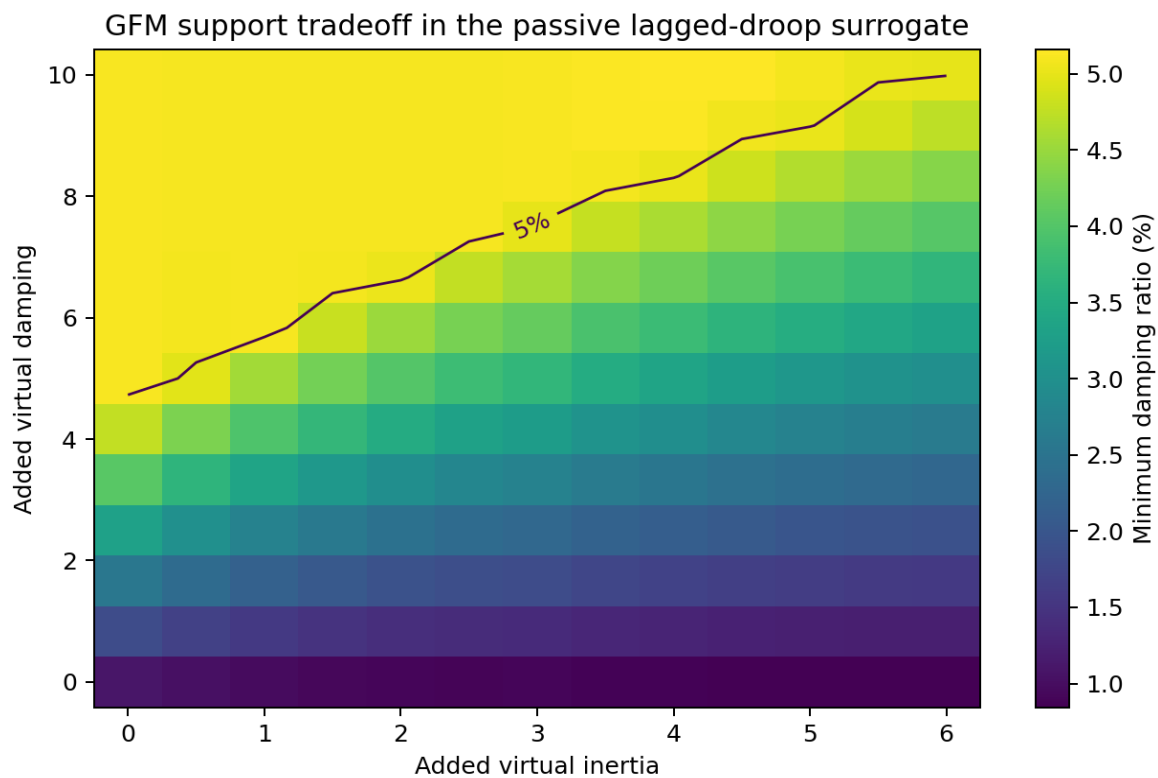


Fig. 10. Virtual inertia and damping are not interchangeable: additional inertia generally requires additional damping to maintain the same damping-ratio target in this surrogate.

9. What is genuinely new from this campaign

Result	Status after this sweep
Stable-but-unsafe existence	Supported at mechanism level: zeta_min >= 5% coexists with G_phys up to 32.2 in the tested controller grid.
Dominant-pole non-identifiability of transient	Strong mechanism counterexample: two cases with nearly identical critical pole and zeta differ by 4.54x in G_phys.
Renyi/free-energy bridge	Useful: beta->0 average-channel behavior can remain similar while beta->infinity worst-direction amplification.
Gauge/coordinate robustness	The physical input-output gain is invariant under consistent similarity transformations by construction; a 12-dec
Cycle pressure as transient predictor	Not supported as a standalone ordering metric in this surrogate. It is a separate interaction-complexity object;
Isolated-mode alpha^1/2 scaling extension	Not quantitatively reproduced for the new controller-aware damping-action Q. Claim must remain limited to the
Topology-induced nonmodal Braess reversal	Not found in the tested line-strength campaign. Standardized line 25-38 reduced G_phys.

10. Next falsification gate

The result worth pursuing now is narrower and stronger than "eigenvalues fail": in a controller-coupled IBR surrogate, dominant modal margins can be nearly identical while the physically normalized pre-asymptotic gain differs by several-fold. The decisive next experiment is to reproduce this matched-pole counterexample in a documented full SG/GFL/GFM linearization and then apply the worst-case initial direction in nonlinear time-domain simulation with current limits and PLL protection. Only that step can upgrade "large physical gain" to a field-relevant protection-activation claim. The gauge/non-Abelian part also requires a genuinely multivariable dq/PQ port model; the present active-channel surrogate cannot establish non-Abelian cycle causation by itself.

Recommended paper framing if full-order validation succeeds: "Stable but Unsafe: Gauge-Invariant Cycle-Resolved Nonmodal Amplification in Inverter-Dominated Power Grids." The headline figure should be the matched-dominant-pole pair, not a generic heatmap.

Reproducibility package

The accompanying folder contains raw CSVs for all sweeps, the case39 input, figures, summary.json, and reproduce_sweeps.py. The script compiles and uses only NumPy, SciPy, pandas, NetworkX, and the local MATPOWER case file.